

C Programming Coding Practice Set – 01

(Covers all topics learned so far: Variables, Data Types, Constants, Input/Output, Operators, Control Instructions, Loops, and Functions)

Level 1: Basic Programs (Input/Output + Variables)

1. Print "Hello World".

2. Input your name and age and display them.

Output Example:

```
Enter name: Gaurav
Enter age: 20
```

```
Name = Gaurav
Age = 20
```

3. Input two numbers and print their sum.

4. Input length and breadth of a rectangle and find area.

5. Input radius and calculate area of a circle.

Use:

```
const float PI = 3.14;
```

Level 2: Operators

6. Input two numbers and perform:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus

7. Find quotient and remainder.

Example:

$$17 \div 5$$

Quotient = 3

Remainder = 2

8. Swap two numbers using a third variable.

9. Swap two numbers without using a third variable.

10. Increment and decrement a variable using ++ and --.

Level 3: Decision Control (if-else)

11. Check whether a number is positive or negative.

12. Check whether a number is even or odd.

13. Find the largest of two numbers.

14. Find the largest among three numbers.

15. Check whether a year is leap year or not.

16. Check whether a person is eligible to vote.

(Age $\geq$ 18)

17. Find absolute value of a number.

Example:

Input = -7

Output = 7

Level 4: Ternary Operator

18. Find greater between two numbers using ? :

19. Check even or odd using ternary operator.

20. Find maximum among two numbers using ternary operator.

Level 5: Switch Case

21. Create a simple calculator using switch.

Operations:

+

-

*

/

%

22. Display day name according to day number (1–7).

Example:

1 → Monday

2 → Tuesday

23. Display month name according to month number.

Level 6: While Loop

24. Print numbers from 1 to 10.

25. Print numbers from 10 to 1.

26. Print even numbers from 1 to 100.

27. Find sum of first N natural numbers.

28. Find factorial of a number.

Example:

5! = 120

29. Reverse a number.

Example:

Input = 1234

Output = 4321

30. Count digits in a number.

Example:

12345 → 5 digits

Level 7: For Loop

31. Print multiplication table of a number.

Example:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

...

$$5 \times 10 = 50$$

32. Find sum of odd numbers from 1 to N.

33. Find sum of even numbers from 1 to N.

34. Find power of a number.

Example:

$$2^5 = 32$$

35. Check whether a number is prime or not.

36. Check whether a number is palindrome or not.

Example:

121 → Palindrome

123 → Not Palindrome

37. Find the sum of digits.

Example:

1234

$1+2+3+4=10$

Level 8: Pattern Programs

38.

```
*  
**  
***  
****  
*****
```

39.

```
1  
12  
123  
1234  
12345
```

40.

```
*****  
****  
***  
**  
*
```

Level 9: Functions

41. Create a function to add two numbers.

42. Create a function to find square of a number.

43. Create a function to check whether a number is even or odd.

44. Create a function to find factorial.

45. Create a function to find maximum between two numbers.

46. Create a function to find area of a circle.

Use:

```
const float PI = 3.14;
```

47. Create a function to reverse a number.

48. Create a function to count digits in a number.

49. Create a function to check prime number.

50. Create a function to check palindrome number.